# FlexGroup 卷的灾难恢复

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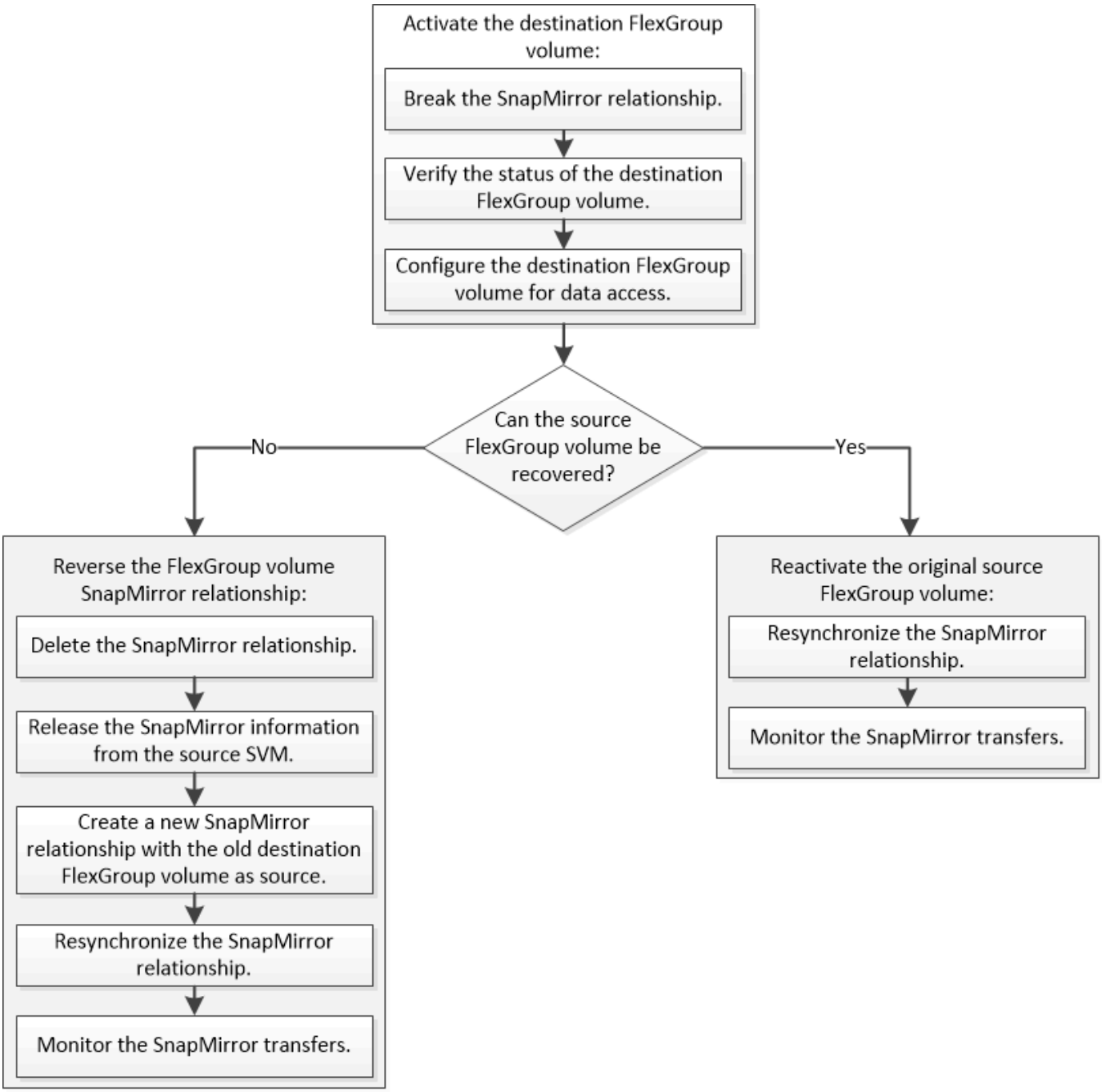
# FlexGroup 卷的灾难恢复

## FlexGroup 卷的灾难恢复工作流

当源 FlexGroup 卷发生灾难时，您应激活目标 FlexGroup 卷并重定向客户端访问。根据源 FlexGroup 卷是否可以恢复，您应重新激活源 FlexGroup 卷或反转 SnapMirror 关系。




当某些 SnapMirror 操作（例如 SnapMirror 中断和重新同步）正在运行时，客户端对目标 FlexGroup 卷的访问会暂时被阻止。如果 SnapMirror 操作失败，则某些成分卷可能仍处于此状态，并且对 FlexGroup 卷的访问被拒绝。在这种情况下，您必须重试 SnapMirror 操作。

# 激活目标 FlexGroup 卷

如果源 FlexGroup 卷由于数据损坏，意外删除或脱机状态等事件而无法提供数据，则必须激活目标 FlexGroup 卷以提供数据访问，直到恢复源 FlexGroup 卷上的数据为止。激活涉及停止未来的 SnapMirror 数据传输并中断 SnapMirror 关系。

您必须从目标集群执行此任务。

步骤

1. 禁用 FlexGroup 卷 SnapMirror 关系的未来传输： `snapmirror quiesce dest_sVM ： dest_flexgroup`

   ```
   cluster2::> snapmirror quiesce -destination-path vsd:dst
   ```

2. 断开 FlexGroup 卷 SnapMirror 关系： `snapmirror break dest_sVM ： dest_flexgroup`

   ```
   cluster2::> snapmirror break -destination-path vsd:dst
   ```

3. 查看 SnapMirror 关系的状态： `snapmirror show -expand`

```
cluster2::> snapmirror show -expand

Progress
Source            Destination Mirror  Relationship   Total
Last
Path        Type  Path        State   Status         Progress  Healthy
Updated
----------- ---- ------------ ------- -------------- --------- -------
--------
vss:s       XDP  vsd:dst        Broken-off
                                    Idle           -         true    -
vss:s__0001 XDP  vsd:dst__0001  Broken-off
                                    Idle           -         true    -
vss:s__0002 XDP  vsd:dst__0002  Broken-off
                                    Idle           -         true    -
vss:s__0003 XDP  vsd:dst__0003  Broken-off
                                    Idle           -         true    -
vss:s__0004 XDP  vsd:dst__0004  Broken-off
                                    Idle           -         true    -
vss:s__0005 XDP  vsd:dst__0005  Broken-off
                                    Idle           -         true    -
vss:s__0006 XDP  vsd:dst__0006  Broken-off
                                    Idle           -         true    -
vss:s__0007 XDP  vsd:dst__0007  Broken-off
                                    Idle           -         true    -
vss:s__0008 XDP  vsd:dst__0008  Broken-off
                                    Idle           -         true    -
...
```

每个成分卷的 SnapMirror 关系状态为 `broken-off` 。

4. 验证目标 FlexGroup 卷是否为读 / 写卷： `volume show -vserver svm_name`

```
cluster2::> volume show -vserver vsd
Vserver   Volume       Aggregate    State      Type       Size
Available Used%
--------- ------------ ------------ ---------- ---- ----------
---------- -----
vsd       dst          -            online     **RW**          2GB
1.54GB   22%
vsd       d2           -            online     DP          2GB
1.55GB   22%
vsd       root_vs0     aggr1        online     RW        100MB
94.02MB    5%
3 entries were displayed.
```

5. 将客户端重定向到目标 FlexGroup 卷。

# 发生灾难后重新激活原始源 FlexGroup 卷

当源 FlexGroup 卷变为可用时，您可以重新同步原始源 FlexGroup 卷和原始目标 卷。目标 FlexGroup 卷上的任何新数据将丢失。

在执行重新同步之前，目标卷上的任何活动配额规则都将被停用，并且配额规则将被删除。

重新同步操作完成后，您可以使用 `volume quota policy rule create` 和 `volume quota modify` 命令创建和重新激活配额规则。

步骤

1. 从目标集群重新同步 FlexGroup 卷 SnapMirror 关系： `snapmirror resync -destination-path dst_sVM ： dest_flexgroup`
2. 查看 SnapMirror 关系的状态： `snapmirror show -expand`

```
cluster2::> snapmirror show -expand

Progress
Source            Destination Mirror  Relationship   Total
Last
Path        Type  Path        State   Status         Progress  Healthy
Updated
----------- ---- ------------ ------- -------------- --------- -------
--------
vss:s       XDP  vsd:dst        Snapmirrored
                                      Idle           -         true    -
vss:s__0001 XDP  vsd:dst__0001  Snapmirrored
                                      Idle           -         true    -
vss:s__0002 XDP  vsd:dst__0002  Snapmirrored
                                      Idle           -         true    -
vss:s__0003 XDP  vsd:dst__0003  Snapmirrored
                                      Idle           -         true    -
vss:s__0004 XDP  vsd:dst__0004  Snapmirrored
                                      Idle           -         true    -
vss:s__0005 XDP  vsd:dst__0005  Snapmirrored
                                      Idle           -         true    -
vss:s__0006 XDP  vsd:dst__0006  Snapmirrored
                                      Idle           -         true    -
vss:s__0007 XDP  vsd:dst__0007  Snapmirrored
                                      Idle           -         true    -
vss:s__0008 XDP  vsd:dst__0008  Snapmirrored
                                      Idle           -         true    -
...
```

每个成分卷的 SnapMirror 关系状态为 `Snapmirrored` 。

# 在灾难恢复期间反转 FlexGroup 卷之间的 SnapMirror 关系

如果灾难禁用了 SnapMirror 关系中的源 FlexGroup 卷，则可以在修复或替换源 FlexGroup 卷时使用目标 FlexGroup 卷提供数据。源 FlexGroup 卷联机后，您可以将原始源 FlexGroup 卷设为只读目标，并反转 SnapMirror 关系。

在执行重新同步之前，目标卷上的任何活动配额规则都将被停用，并且配额规则将被删除。

重新同步操作完成后，您可以使用 `volume quota policy rule create` 和 `volume quota modify` 命令创建和重新激活配额规则。

步骤

1. 在初始目标 FlexGroup 卷上，删除源 FlexGroup 卷与目标 FlexGroup 卷之间的数据保护镜像关系：`snapmirror delete -destination-path svm_name： volume_name`

```
cluster2::> snapmirror delete -destination-path vsd:dst
```

2. 在原始源 FlexGroup 卷上，从源 FlexGroup 卷中删除关系信息： `snapmirror release -destination -path svm_name ： volume_name -relationship-info-only`

   删除 SnapMirror 关系后，必须先从源 FlexGroup 卷中删除关系信息，然后再尝试重新同步操作。

```
cluster1::> snapmirror release -destination-path vsd:dst -relationship
-info-only true
```

3. 在新的目标 FlexGroup 卷上，创建镜像关系： `snapmirror create -source-path src_svm_name ： volume_name -destination-path dst_svm_name ： volume_name -type XDP -policy MirrorAllSnapshots`

```
cluster1::> snapmirror create -source-path vsd:dst -destination-path
vss:src -type XDP -policy MirrorAllSnapshots
```

4. 在新的目标 FlexGroup 卷上，重新同步源 FlexGroup： `snapmirror resync -source-path svm_name ： volume_name`

```
cluster1::> snapmirror resync -source-path vsd:dst
```

5. 监控 SnapMirror 传输： `snapmirror show -expand`

```
cluster2::> snapmirror show -expand

Progress
Source            Destination Mirror  Relationship   Total
Last
Path        Type  Path        State   Status         Progress  Healthy
Updated
----------- ---- ------------ ------- -------------- --------- -------
--------
vsd:dst       XDP  vss:src        Snapmirrored
                                       Idle           -         true    -
vss:dst__0001 XDP  vss:src__0001  Snapmirrored
                                       Idle           -         true    -
vsd:dst__0002 XDP  vss:src__0002  Snapmirrored
                                       Idle           -         true    -
vsd:dst__0003 XDP  vss:src__0003  Snapmirrored
                                       Idle           -         true    -
vsd:dst__0004 XDP  vss:src__0004  Snapmirrored
                                       Idle           -         true    -
vsd:dst__0005 XDP  vss:src__0005  Snapmirrored
                                       Idle           -         true    -
vsd:dst__0006 XDP  vss:src__0006  Snapmirrored
                                       Idle           -         true    -
vsd:dst__0007 XDP  vss:src__0007  Snapmirrored
                                       Idle           -         true    -
vsd:dst__0008 XDP  vss:src__0008  Snapmirrored
                                       Idle           -         true    -
...
```

每个成分卷的 SnapMirror 关系状态显示为 `Snapmirrored`，表示重新同步已成功。

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